Midpoint Circle Algorithm

Q. Draw a circle having centre at (0, 0) & radius 10.

Ans:-

We know, if r is integer then,

$$P_0 = 1 - r$$
$$= 1 - 10 = -9$$

and at initial position

$x_0 = 0$ & $y_0 = r$

So, $x_0 = 0$ & $y_0 = 10$

Divide into 4 parts by octa co-ordinates

Conditions:

i) if P < 0, then

$$P_{k+1} = P_k + 2x_{k+1} + 1$$

and (x+1, y)

ii) if P ≥ 0 then

$$P_{k+1} = P_k + 2x_{k+1} + 1 - 2y_{k+1}$$

and (x+1, y−1)

Run until (x's value = y's value)

Iteration Table

k	P_k	x_{k+1}, y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
0	−9	1, 10	2	20
1	−6	2, 10	4	20
2	−1	3, 10	6	20
3	6	4, 9	8	18
4	−3	5, 9	10	18
5	8	6, 8	12	16
6	5	7, 7	14	14

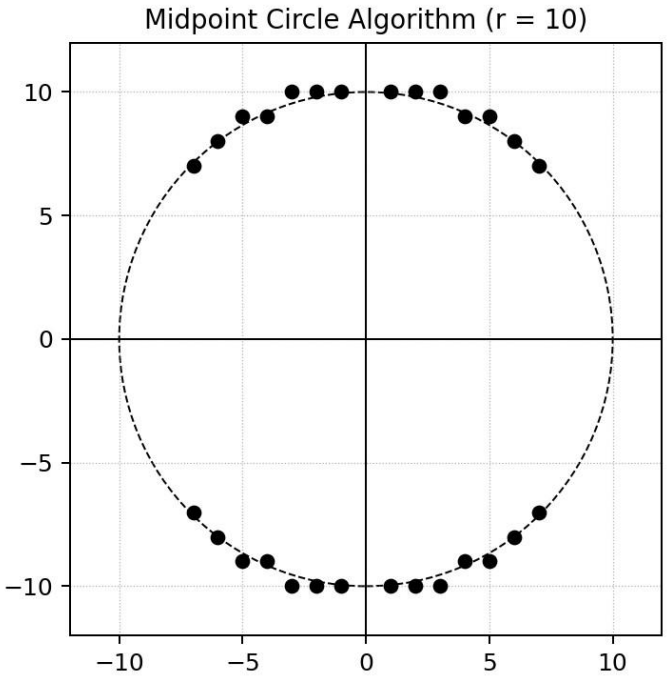
Circle draw → put those in the x & y co-ordinate

$+x\ -y$ $\quad x+\ y+$

$-x\ -y$ $\quad x-\ y+$

Quadrant (Reflection) Table

Q1 (x, y)	Q2 (−x, y)	Q3 (−x, −y)	Q4 (x, −y)
1, 10	-1, 10	-1, -10	1, -10
2, 10	-2, 10	-2, -10	2, -10
3, 10	-3, 10	-3, -10	3, -10
4, 9	-4, 9	-4, -9	4, -9
5, 9	-5, 9	-5, -9	5, -9
6, 8	-6, 8	-6, -8	6, -8
7, 7	-7, 7	-7, -7	7, -7



Plotted points for the midpoint circle algorithm, r = 10, centred at (0, 0).

Bresenham's Circle Drawing Algorithm

→ Point (x, y) = (0, 0)

R = 6

Decision Parameter, (P) = 3 − 2R

$$P_0 = 3 - 2(6) = -9$$

$x_0 = 0$

$y_0 = r = 6$

Divide into 8 parts.

Conditions:

i) if P < 0, then

$$P_{k+1} = P_k + 4x_{k+1} + 6$$

and (x+1, y)

ii) if P ≥ 0 then

$$P_{k+1} = P_k + 4(x_{k+1} - y_{k+1}) + 10$$

and (x+1, y−1)

Run until (x's value = y's value)

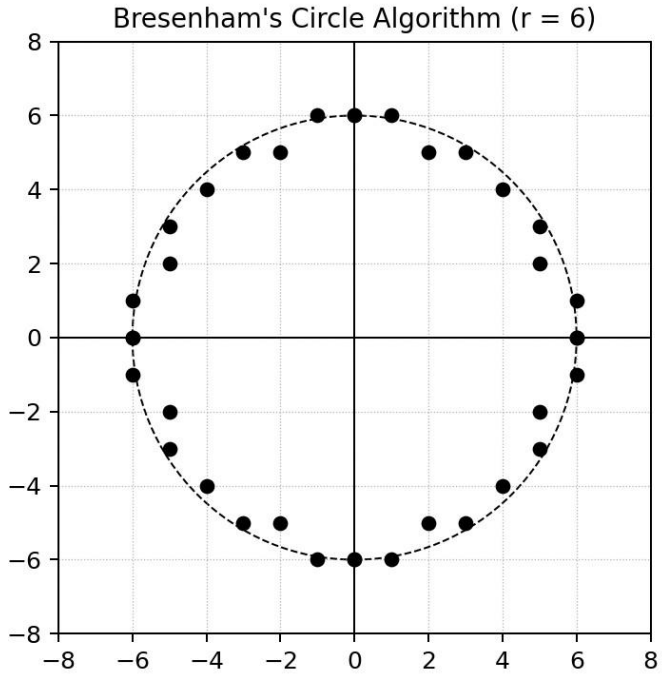
k	P_k	x_k, y_k	x_{k+1}	y_{k+1}
0	−9	0, 6	1	6
1	1	1, 6	2	5
2	−1	2, 5	3	5
3	17	3, 5	4	4

Quad Table

{(5, 3) → from here (mirror & flip)}

Q1 (x, y)	Q2 (−x, y)	Q3 (−x, −y)	Q4 (x, −y)
(0, 6)	(0, 6)	(0,- 6)	(0,- 6)
(1, 6)	(-1, 6)	(-1,- 6)	(1,- 6)
(2, 5)	(-2, 5)	(-2, -5)	(2, -5)
(3, 5)	(-3, 5)	(-3, -5)	(3, -5)
(4, 4)	(-4, 4)	(-4, -4)	(4, -4)
(5, 3)	(-5, 3)	(-5, -3)	(5, -3)
(5, 2)	(-5, 2)	(-5, -2)	(5, -2)
(6, 1)	(-6, 1)	(-6, -1)	(6, -1)
(6, 0)	(-6, 0)	(-6, -0)	(6, -0)

Note: the last row's Q3 and Q4 entries are written as (−6, 0) and (6, 0) rather than "−0", since −0 and 0 are the same value on the y-axis.



Plotted points for Bresenham's circle algorithm, r = 6, centred at (0, 0).